Ex 17

Let $f: \mathbb{R}^d \to \mathbb{R}$ be continously diff. Then the following statments are equivalet.

(i) f is L smooth

(ii) $2D_f(x,y) \le L\|x-y\|^2 \quad \forall x,y \in \mathbb{R}^d$

(iii) $\langle \nabla f(x) - \nabla f(y); x-y \rangle \le L\|x-y\|^2 \quad \forall x,y \in \mathbb{R}^d$

(i) $\Rightarrow$ (ii) $\quad g(x) = \frac{L}{2}\|x\|^2 - f(x)$

If f is L-smooth then $g(x)$ is convex which means that

$$2D_g(x,y) = L\|x\|^2 - L\|y\|^2 - 2f(x) + 2f(y) + 2\langle \nabla f(y); x-y \rangle - 2L\langle y; x-y \rangle \ge 0$$

which mean $\quad 2f(x) - 2f(y) + 2\langle f(y); x-y \rangle \le L\left(\|x\|^2 - \|y\|^2 - 2y; x-y\rangle\right)$

which is the same as $\quad 2D_f(x,y) \le L\|x-y\|^2$ //

(ii) $\overset{\Rightarrow (iii)}{}$ If $2D_f(x,y) \le L\|x-y\|^2$ then $D_g(x,y) \ge 0$ then

$\langle \nabla g(x) - \nabla g(y); x-y \rangle \ge 0$ which is the same as

$L\langle x-y; x-y \rangle - \langle \nabla f(x) - \nabla f(y); x-y \rangle \ge 0$ and as a result

$\langle \nabla f(x) - \nabla f(y); x-y \le L\|x-y\|^2$ //

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(iii) => (i)

If $\langle \nabla f(x) - \nabla f(y), x-y \rangle \leq L\|x-y\|^2$ then

$\langle \nabla g(x) - \nabla g(y), x-y \rangle \geq 0$ which means $g(x)$ is convex

which means that $f(x) = \frac{L}{2}\|x\|^2 - g(x)$ is convex.

Ex[8] If $f: \mathbb{R}^d \to \mathbb{R}$ is $f \in C^2(\mathbb{R}^d)$, then

It is L-smooth iff $\nabla^2 f(x) \preceq LI$

1) If f is L-smooth then $g(x) := \frac{L}{2}\|x\|^2 - f(x)$ is convex and $g(x) \in C^2(\mathbb{R}^d)$ as a result $\nabla^2 g(x) = LI - \nabla^2 f(x) \succeq 0$

as a result $\nabla^2 f(x) \preceq LI$.

2) If $\nabla^2 f(x) \preceq LI$ then $\nabla^2 g(x) \succeq 0$ as a result

$g(x)$ is convex so $f(x)$ is L-smooth.